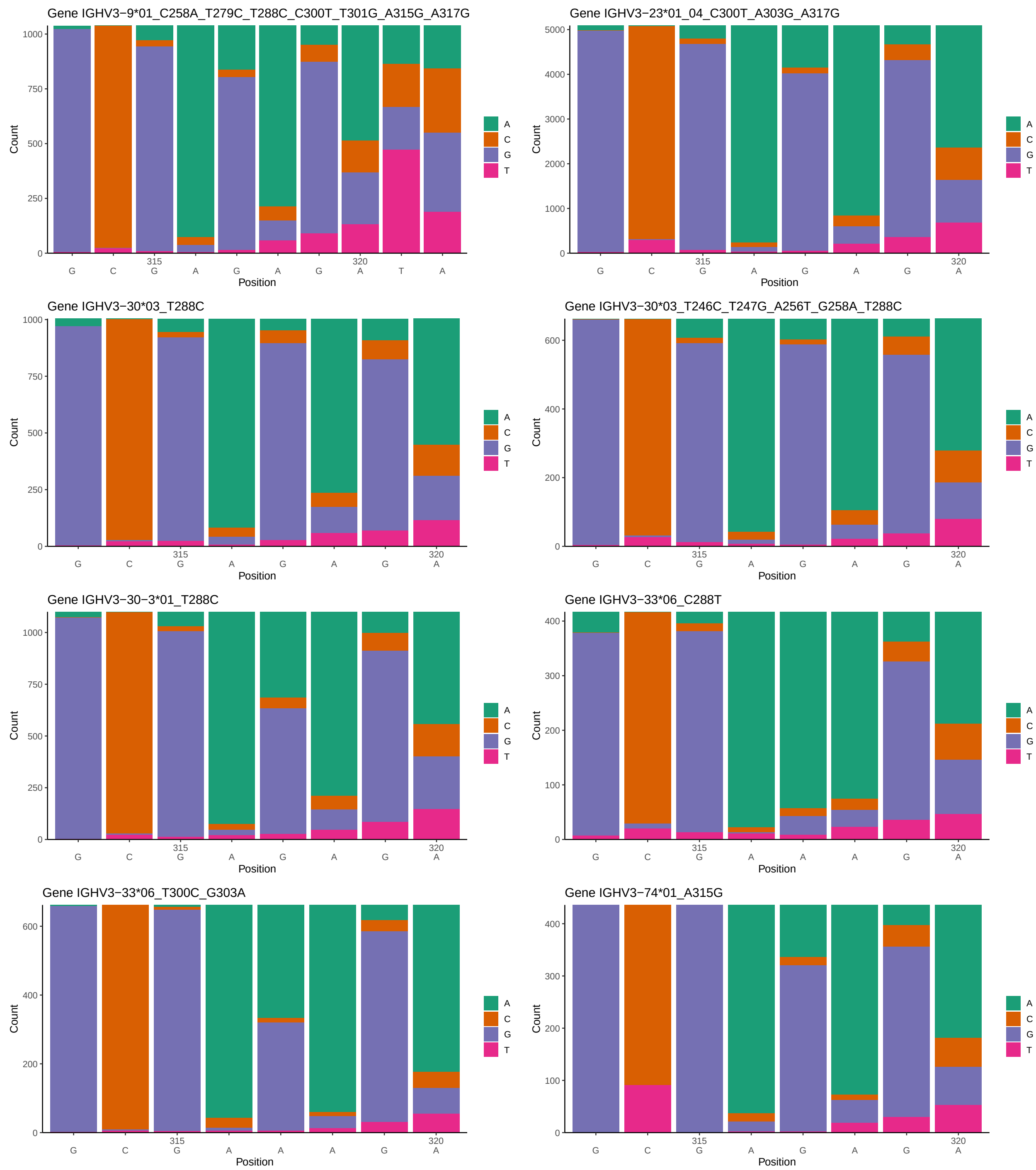
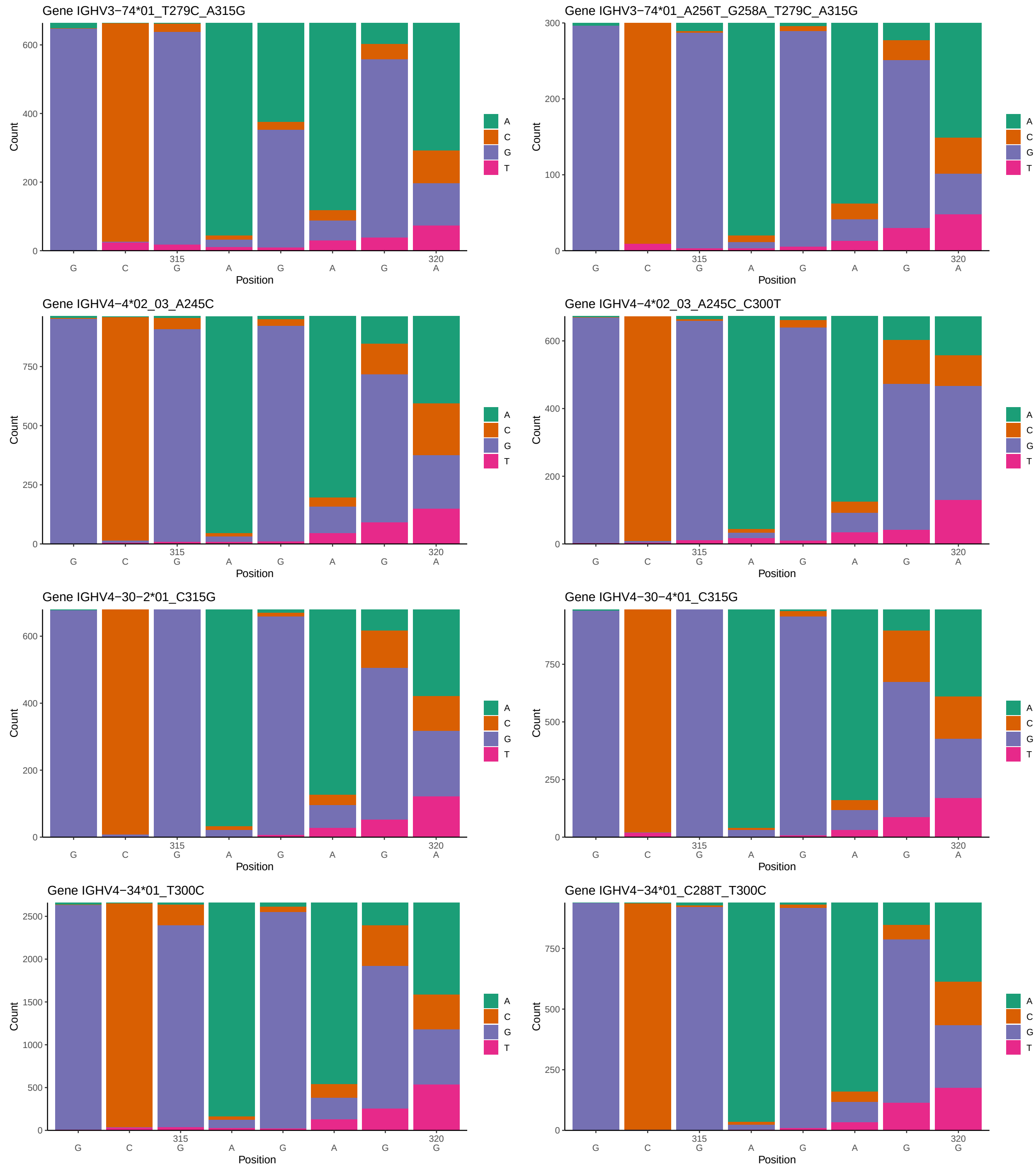
OGRDBstats Report

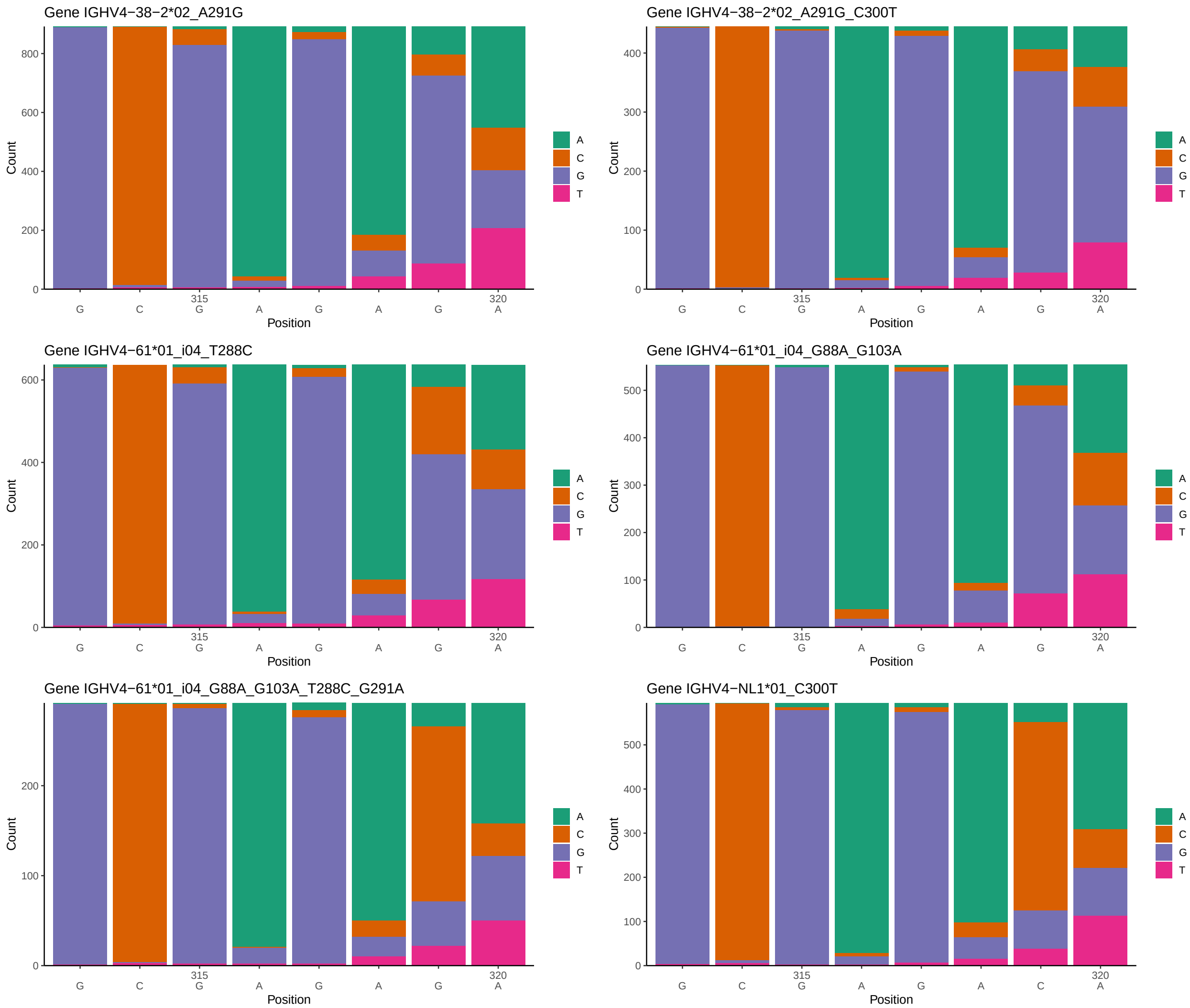
Contents

1 Novel sequence analysis

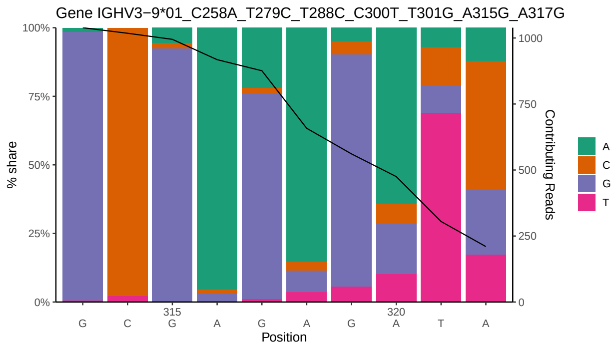
1.1 End-nucleotide composition

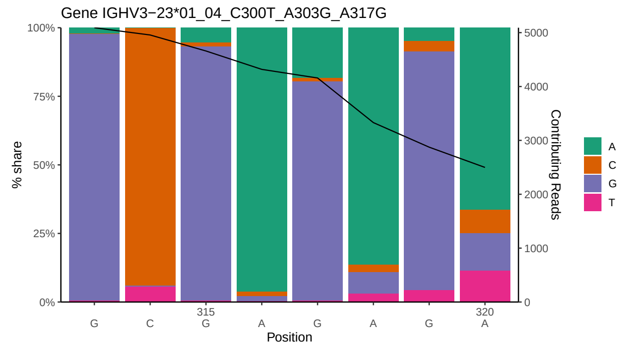


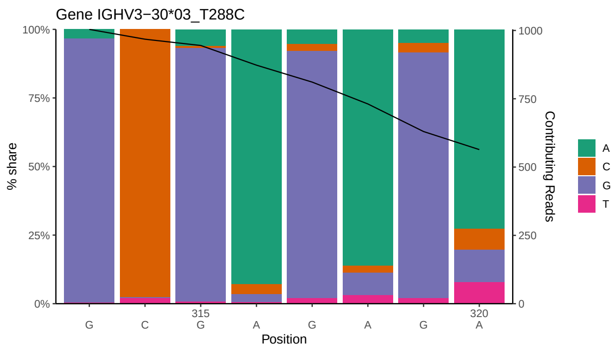

Gene IGHV3−74*01_T279C_A315G
Gene IGHV3−74*01_A256T_G258A_T279C_A315G
Gene IGHV4−4*02_03_A245C
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
Count
Position
G C G A G A G A
315
320
A
C
G
T

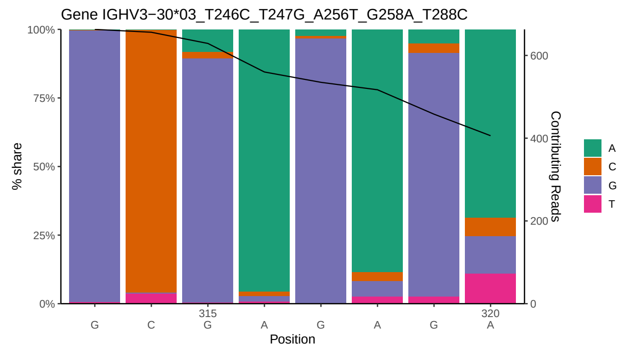

Gene IGHV4−38−2*02_A291G
Gene IGHV4−38−2*02_A291G_C300T
Gene IGHV4−61*01_i04_T288C
Gene IGHV4−61*01_i04_G88A_G103A
Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A
Gene IGHV4−NL1*01_C300T
Count
Position
315
320
G
C
G
A
G
A
G
A
A
C
G
T

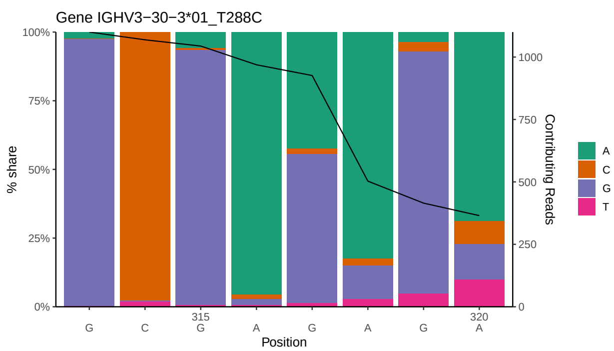
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

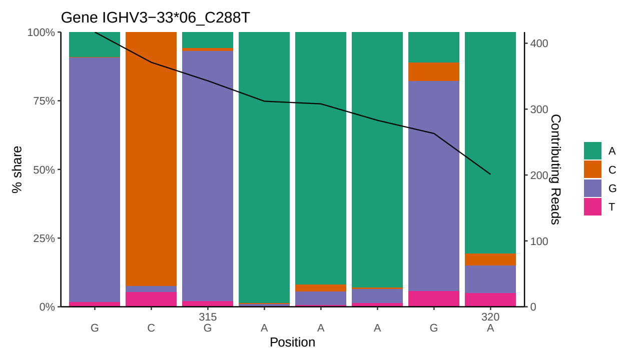


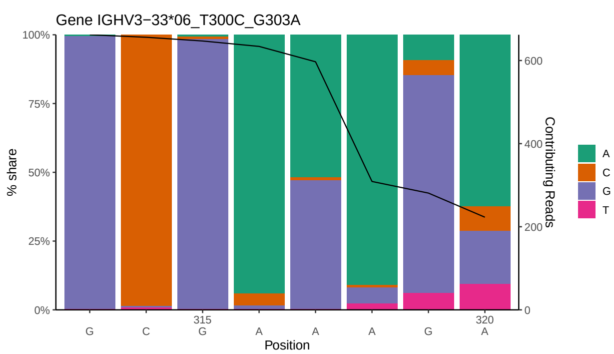


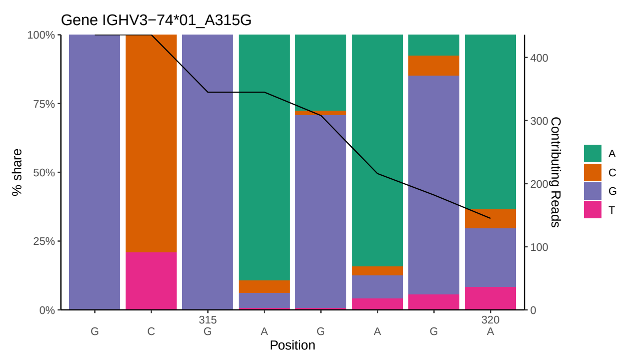


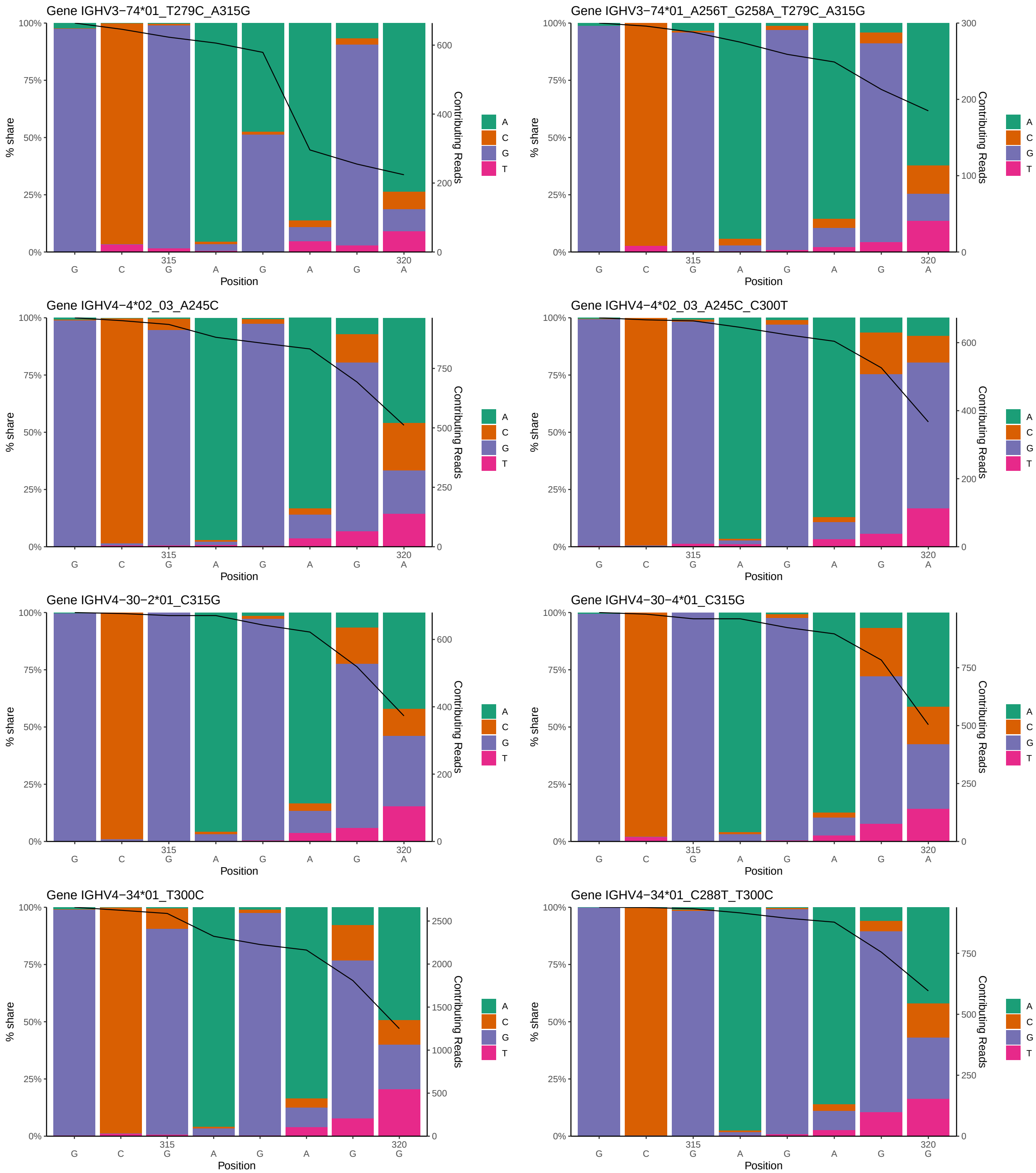


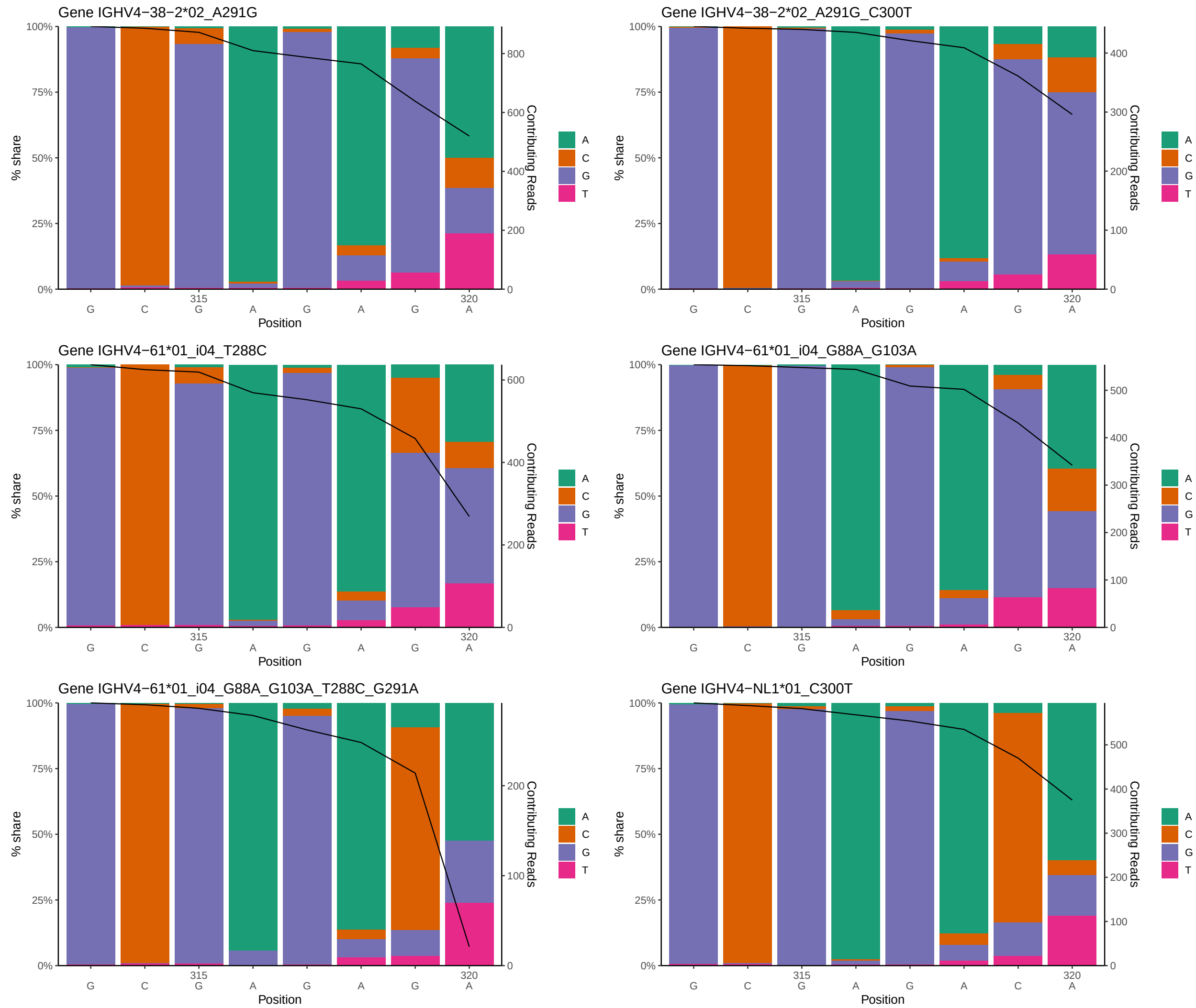




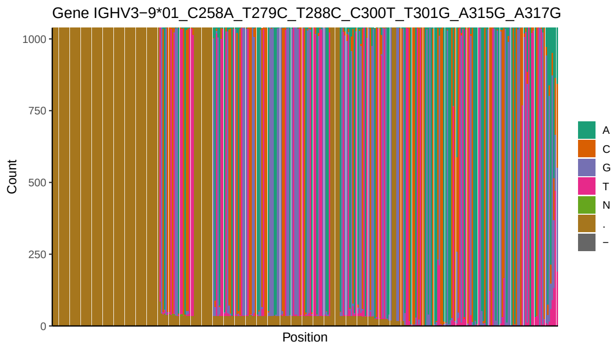


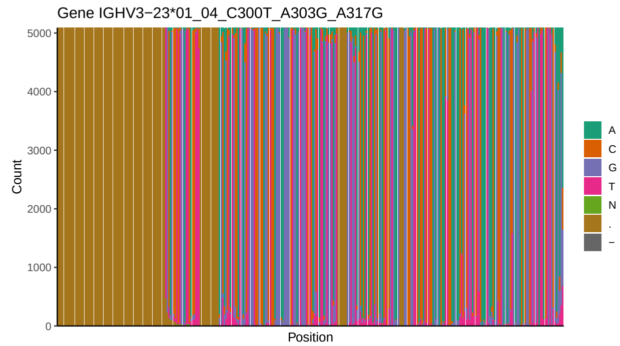


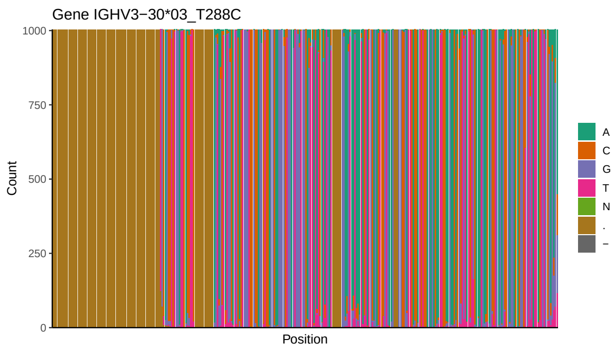

Gene IGHV3−74*01_T279C_A315G
Gene IGHV3−74*01_A256T_G258A_T279C_A315G
Gene IGHV4−4*02_03_A245C
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
% share
Contributing Reads
Position
A
C
G
T

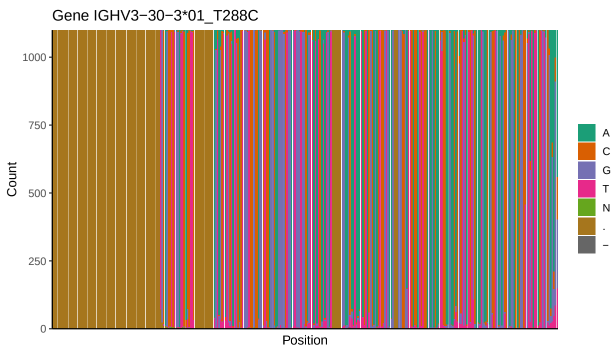

Gene IGHV4−38−2*02_A291G
Gene IGHV4−38−2*02_A291G_C300T
Gene IGHV4−61*01_i04_T288C
Gene IGHV4−61*01_i04_G88A_G103A
Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A
Gene IGHV4−NL1*01_C300T
% share
100%
75%
50%
25%
0%
Contributing Reads
Position
315
320
G C G A G A G A
G C G A G A C A
A
C
G
T

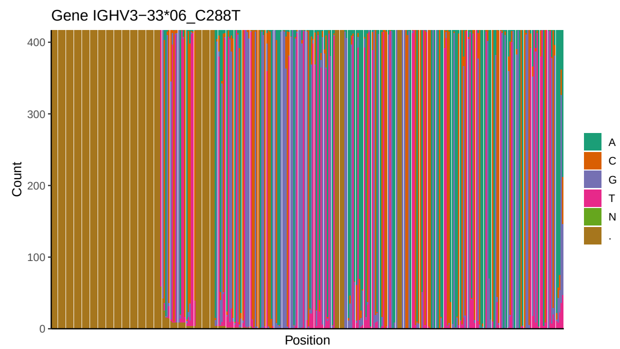
1.3 Whole-sequence composition of each assigned read

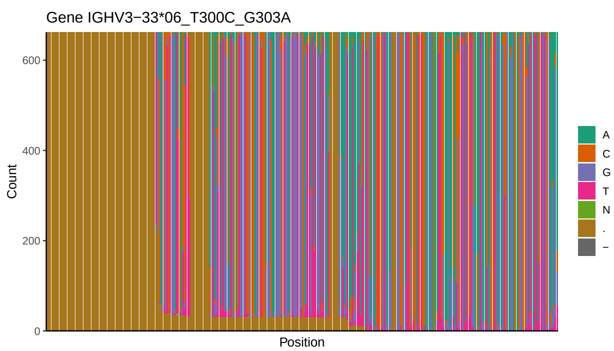




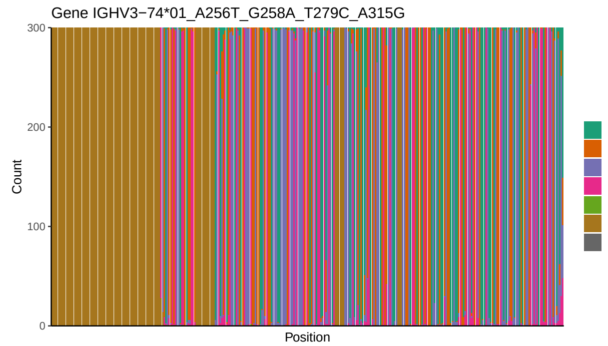


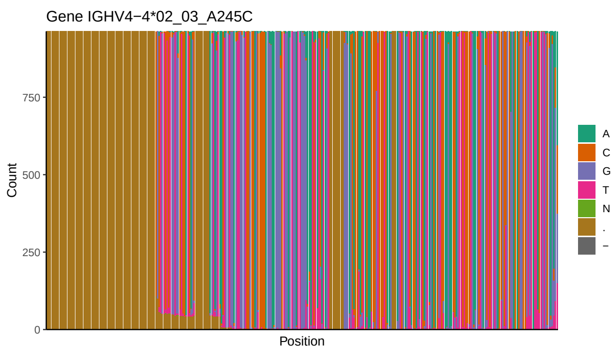


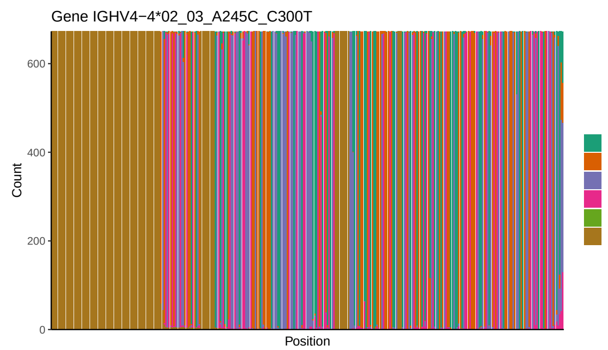


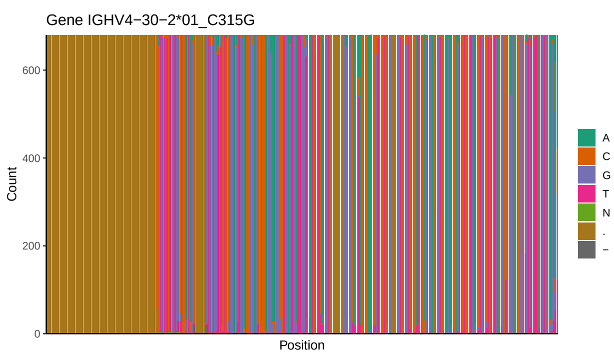


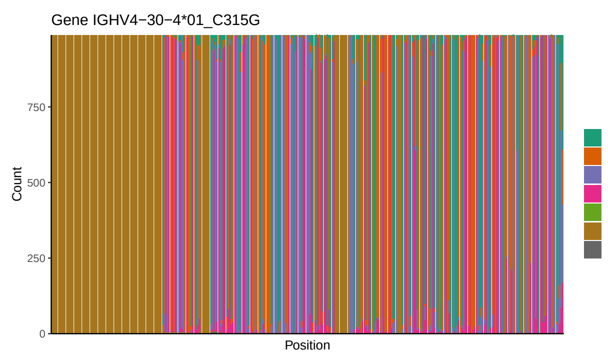

Gene IGHV3−74*01_T279C_A315G
Count
0
200
400
600
Position
A
C
G
T
N
.

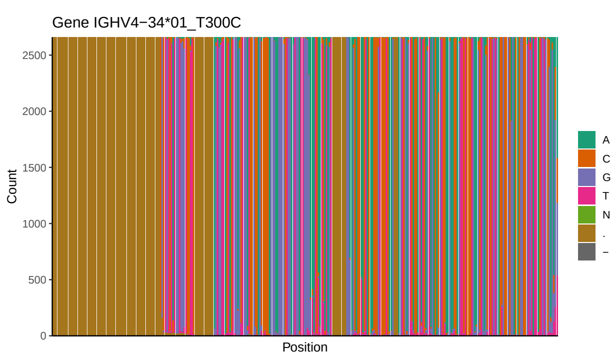

Gene IGHV3−74*01_A256T_G258A_T279C_A315G
Count
0
100
200
300
Position
A
C
G
T
N
.
−

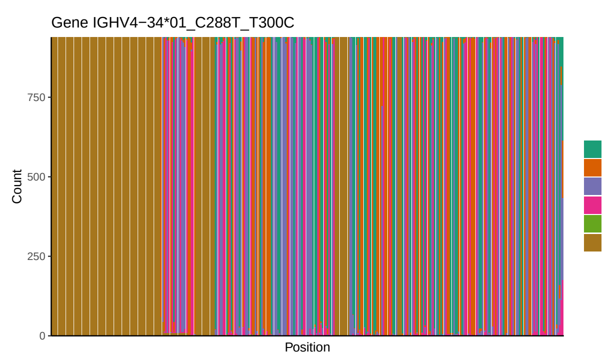

Gene IGHV4−4*02_03_A245C
Count
0
250
500
750
Position
A
C
G
T
N
.
−

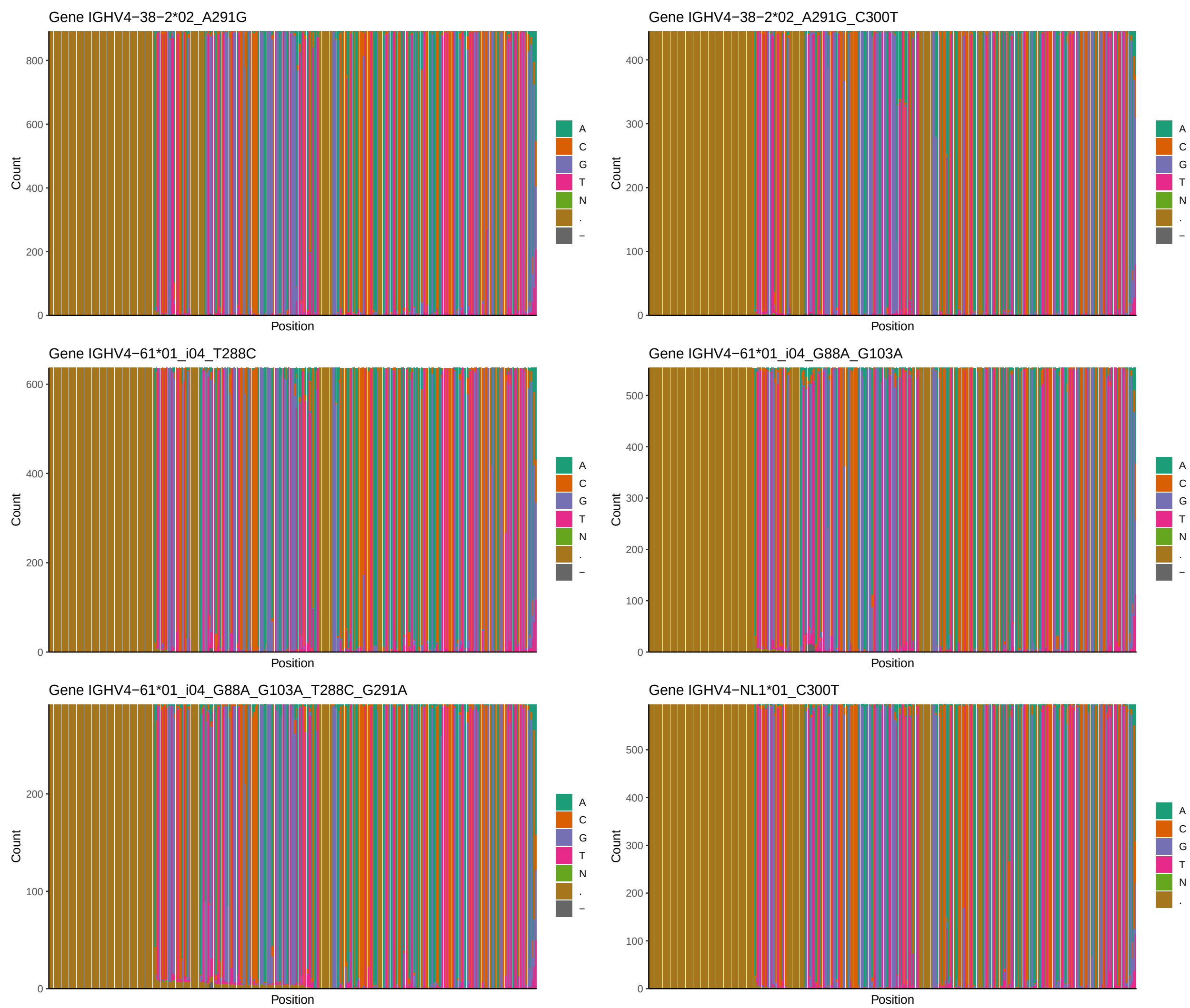

Gene IGHV4−4*02_03_A245C_C300T
Count
0
200
400
600
Position
A
C
G
T
N
.


Gene IGHV4−30−2*01_C315G
Count
0
200
400
600
Position
A
C
G
T
N
.
−

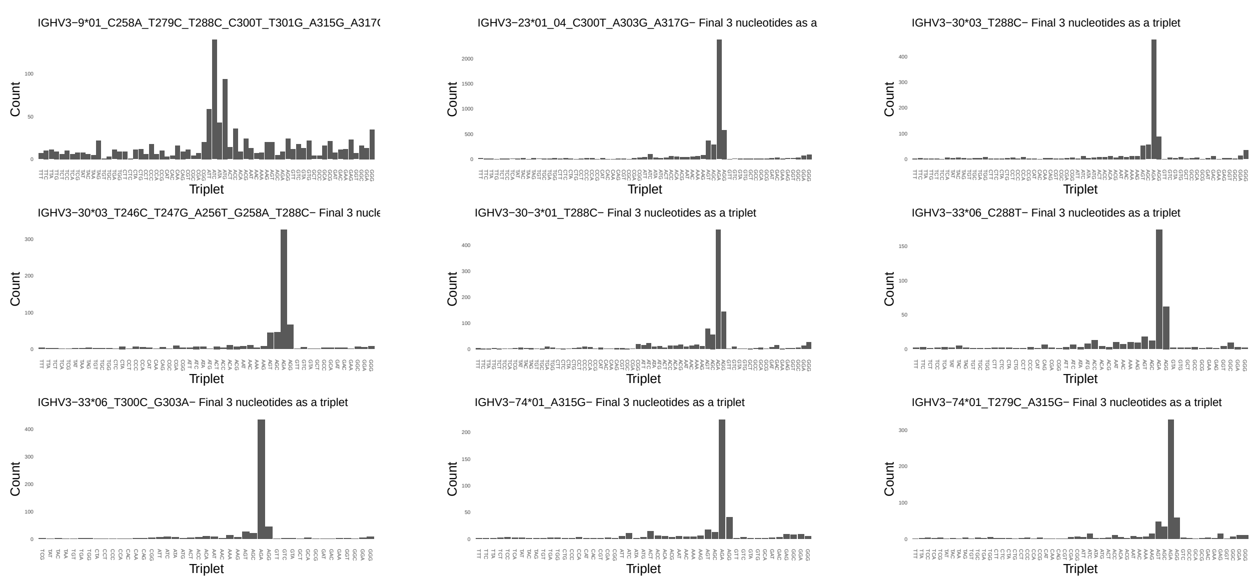

Gene IGHV4−30−4*01_C315G
Count
0
250
500
750
Position
A
C
G
T
N
.
−

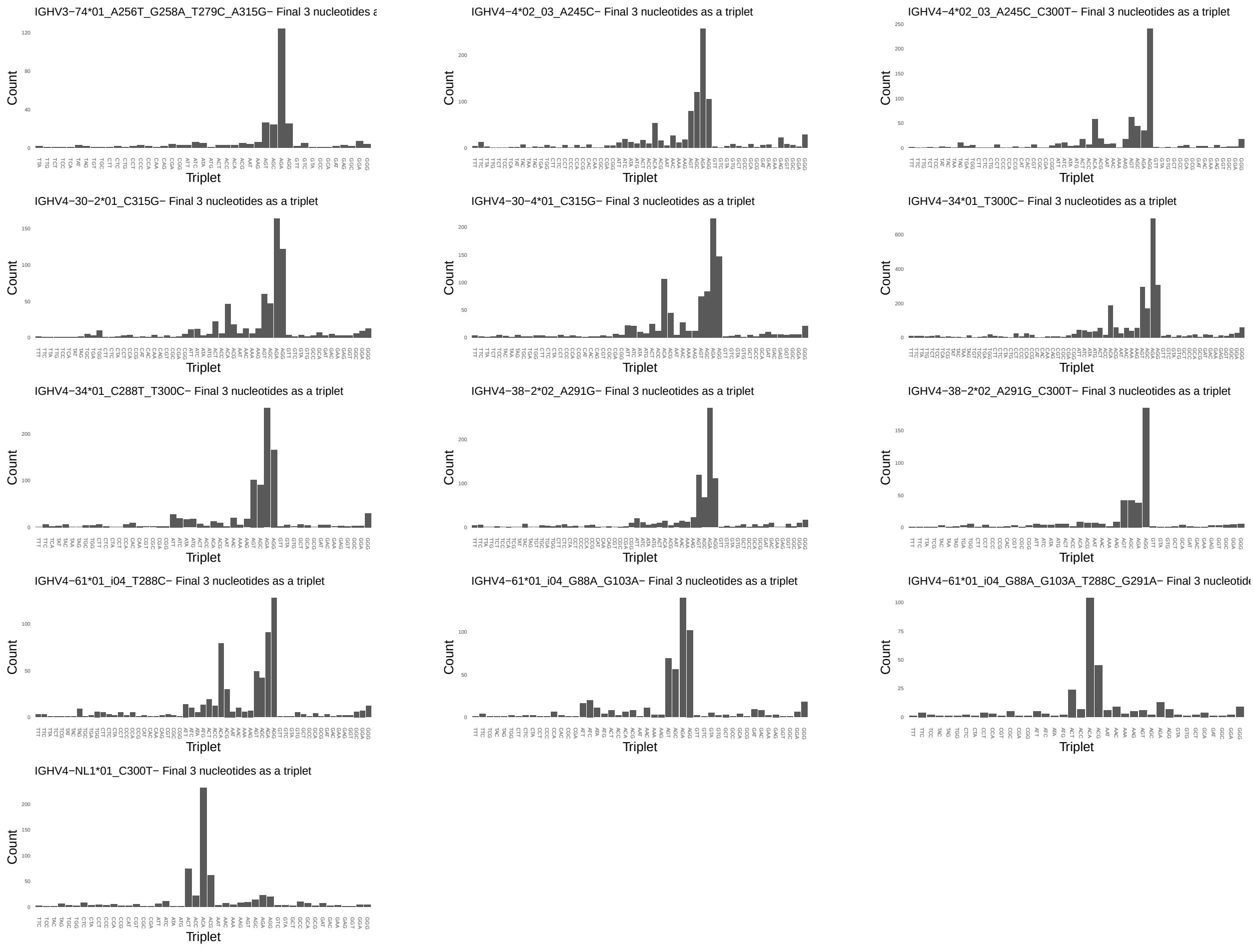

Gene IGHV4−34*01_T300C
Count
0
500
1000
1500
2000
2500
Position
A
C
G
T
N
.
−


Gene IGHV4−34*01_C288T_T300C
Count
0
250
500
750
Position
A
C
G
T
N
.

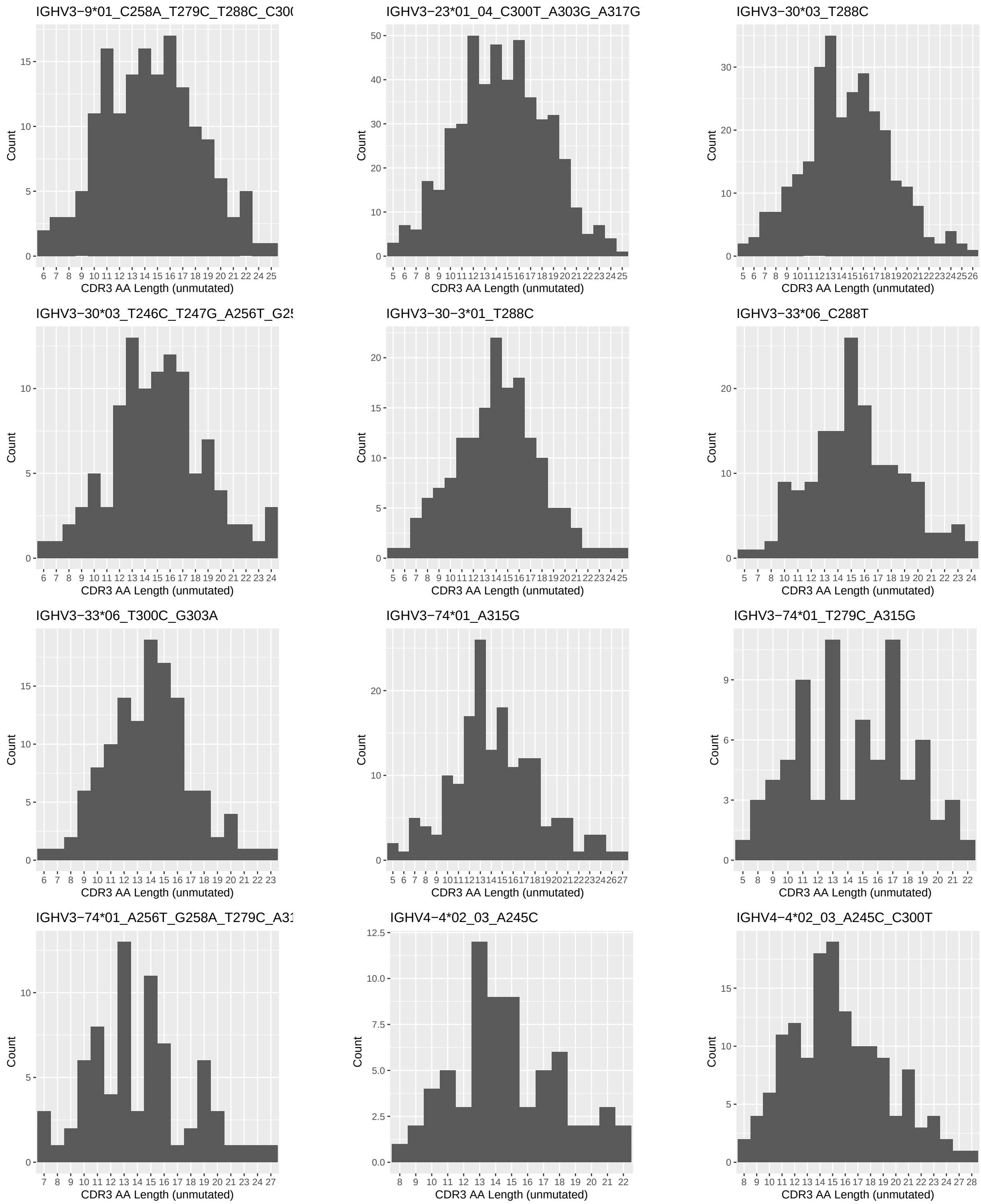


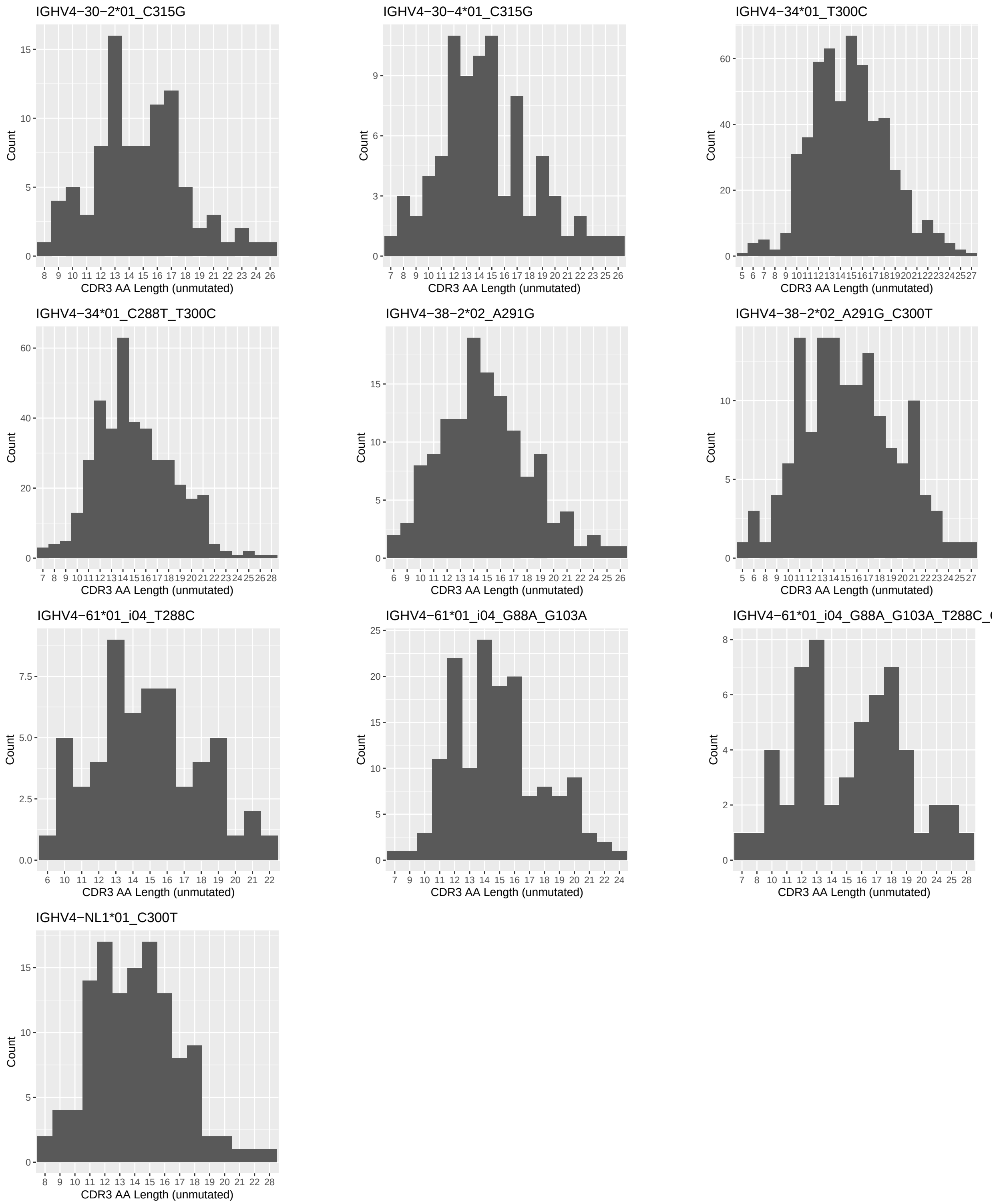
1.4 Final three nucleotides: frequency of each observed triplet



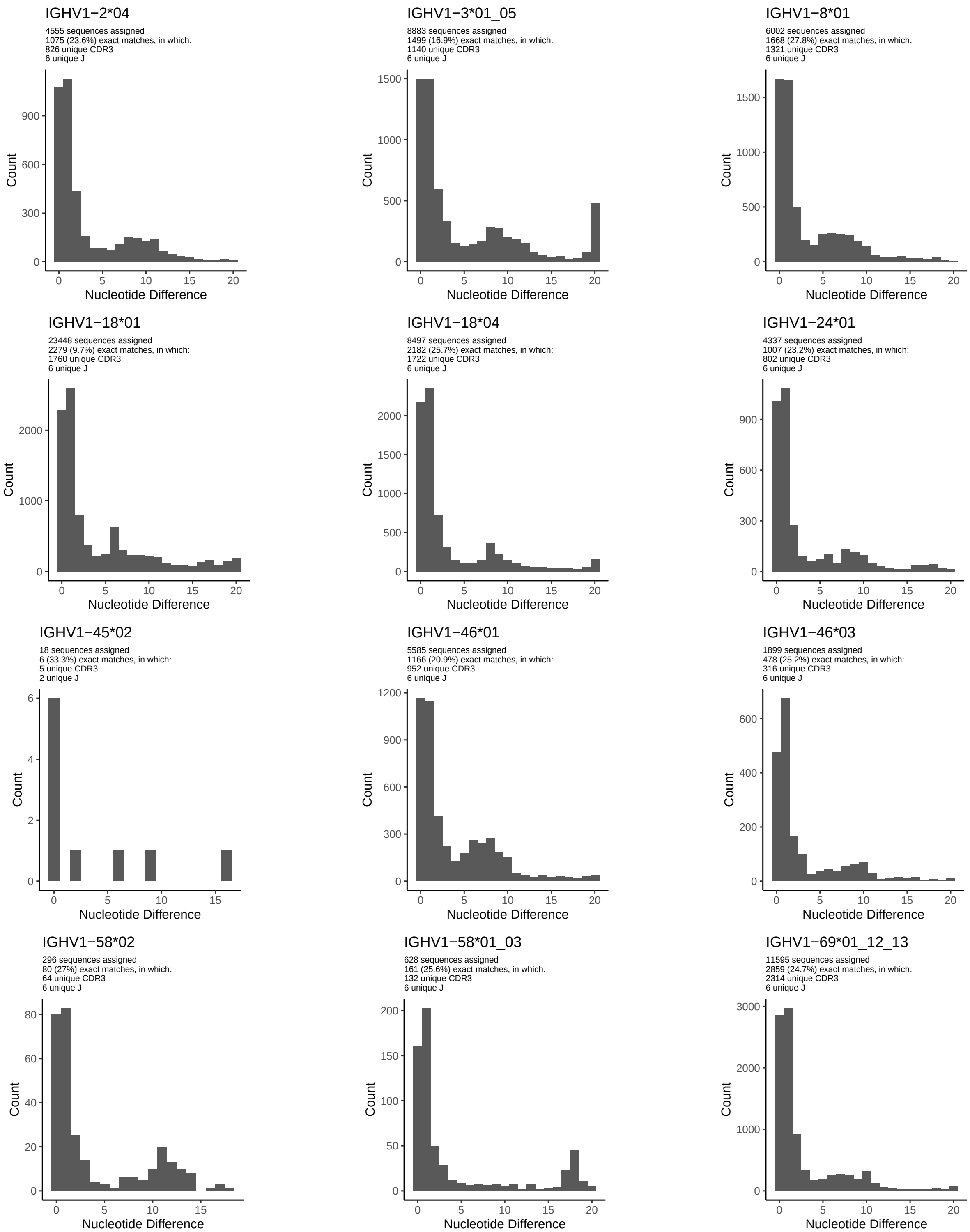

IGHV3−74*01_A256T_G258A_T279C_A315G− Final 3 nucleotides a
IGHV4−4*02_03_A245C− Final 3 nucleotides as a triplet
IGHV4−4*02_03_A245C_C300T− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−4*01_C315G− Final 3 nucleotides as a triplet
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_C288T_T300C− Final 3 nucleotides as a triplet
IGHV4−38−2*02_A291G− Final 3 nucleotides as a triplet
IGHV4−38−2*02_A291G_C300T− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_T288C− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_G103A− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_G103A_T288C_G291A− Final 3 nucleotide
IGHV4−NL1*01_C300T− Final 3 nucleotides as a triplet
Count
Triplet

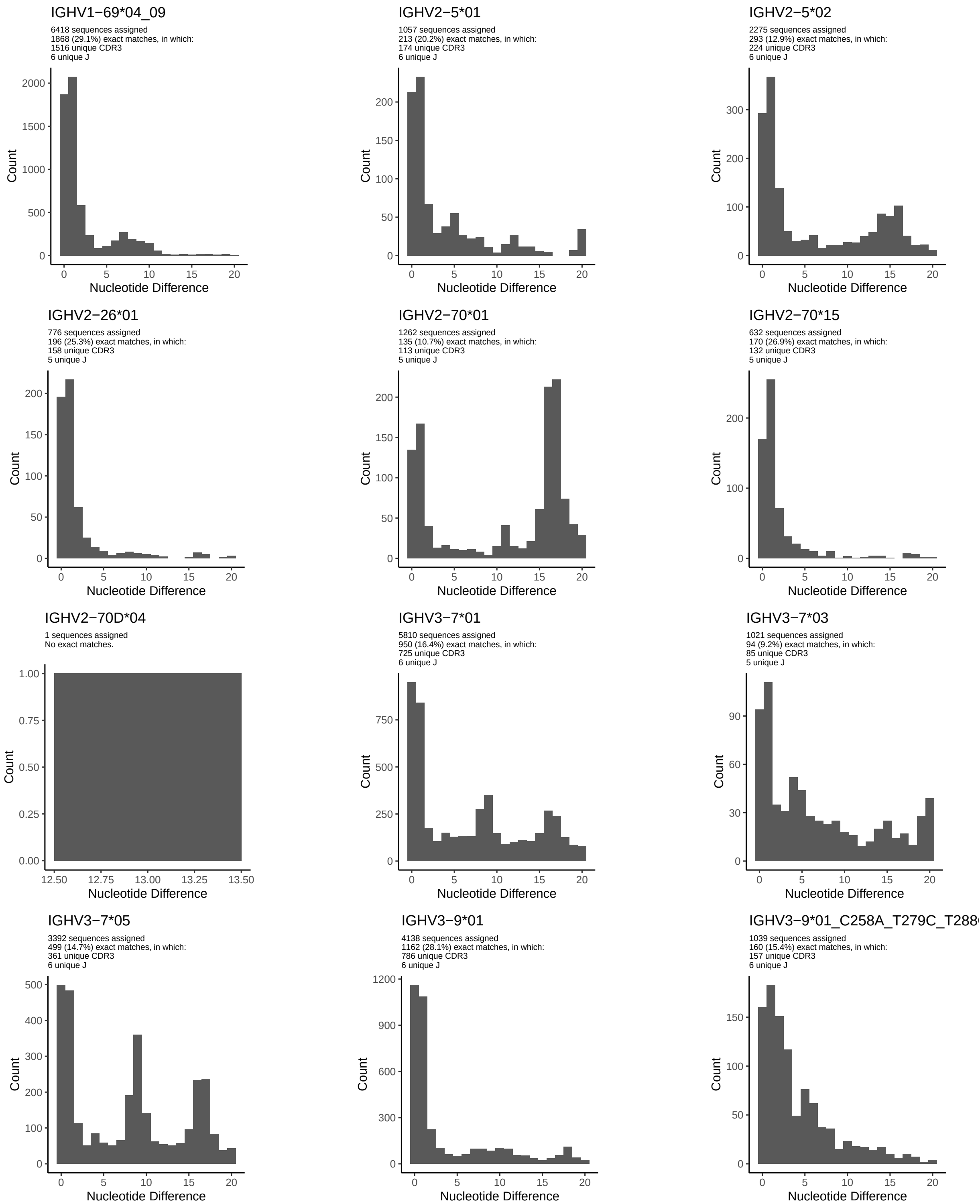
1.5 CDR3 length distribution, in assignments to novel alleles

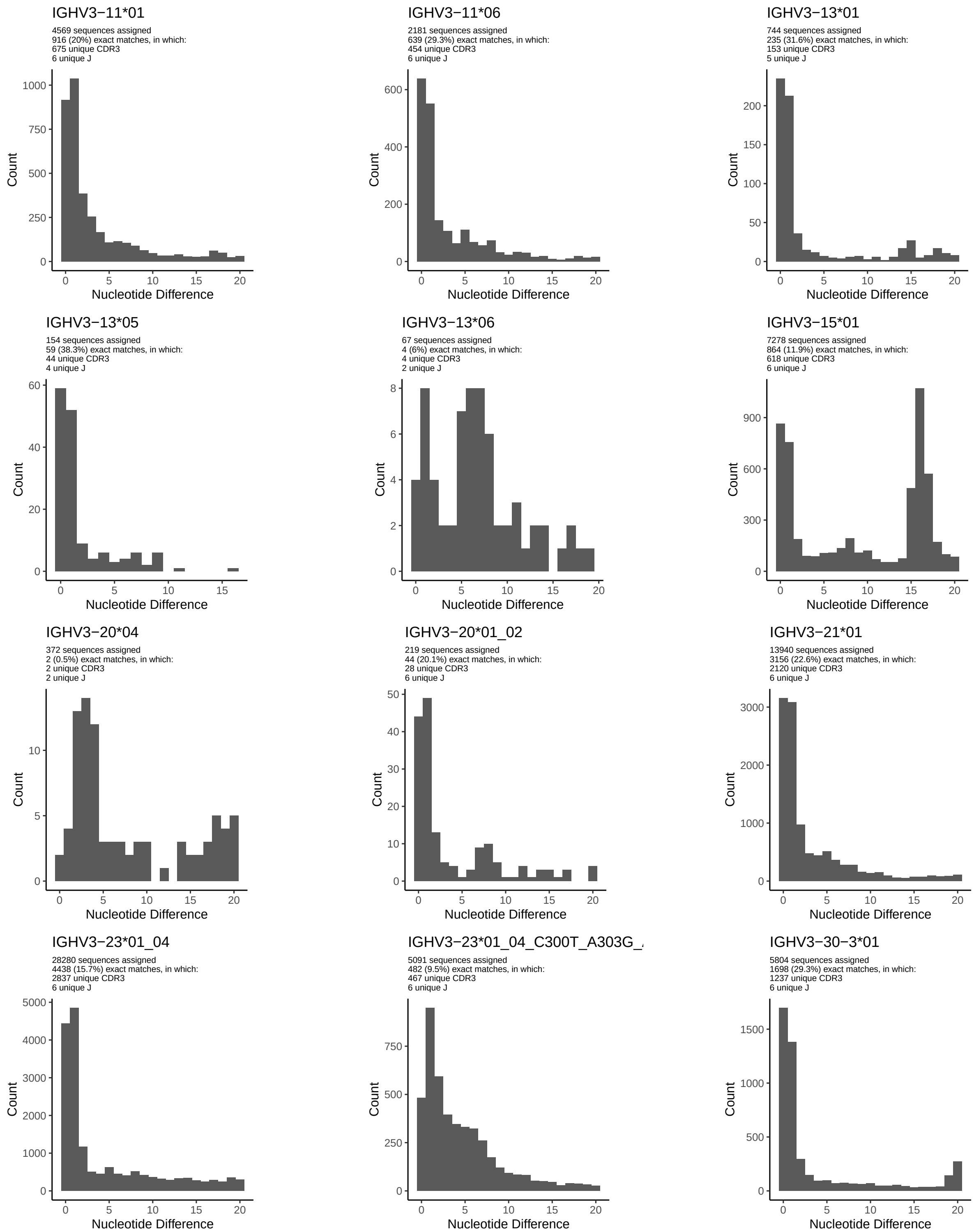


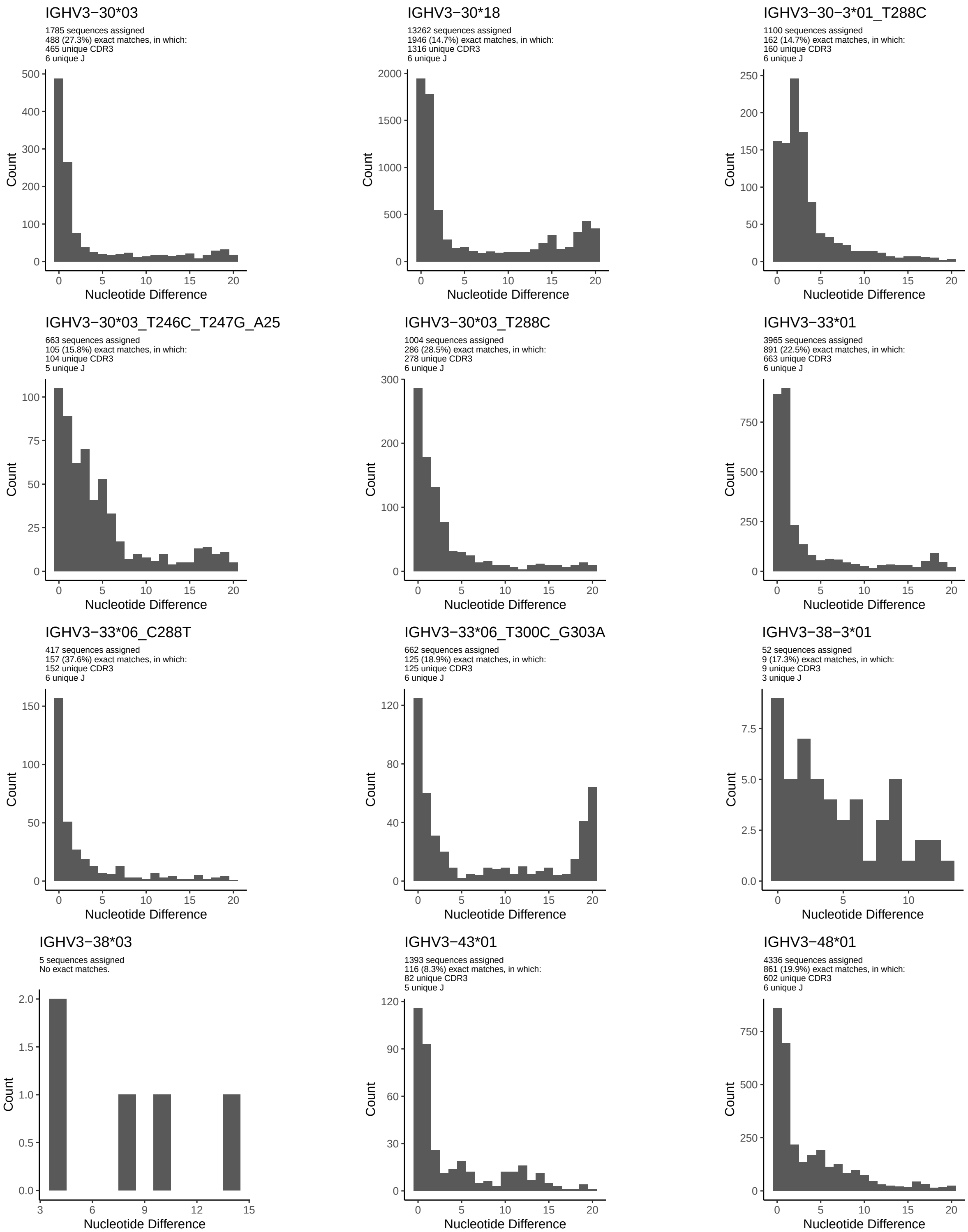

IGHV4−30−2*01_C315G
IGHV4−30−4*01_C315G
IGHV4−34*01_T300C
IGHV4−34*01_C288T_T300C
IGHV4−38−2*02_A291G
IGHV4−38−2*02_A291G_C300T
IGHV4−61*01_i04_T288C
IGHV4−61*01_i04_G88A_G103A
IGHV4−61*01_i04_G88A_G103A_T288C_
IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

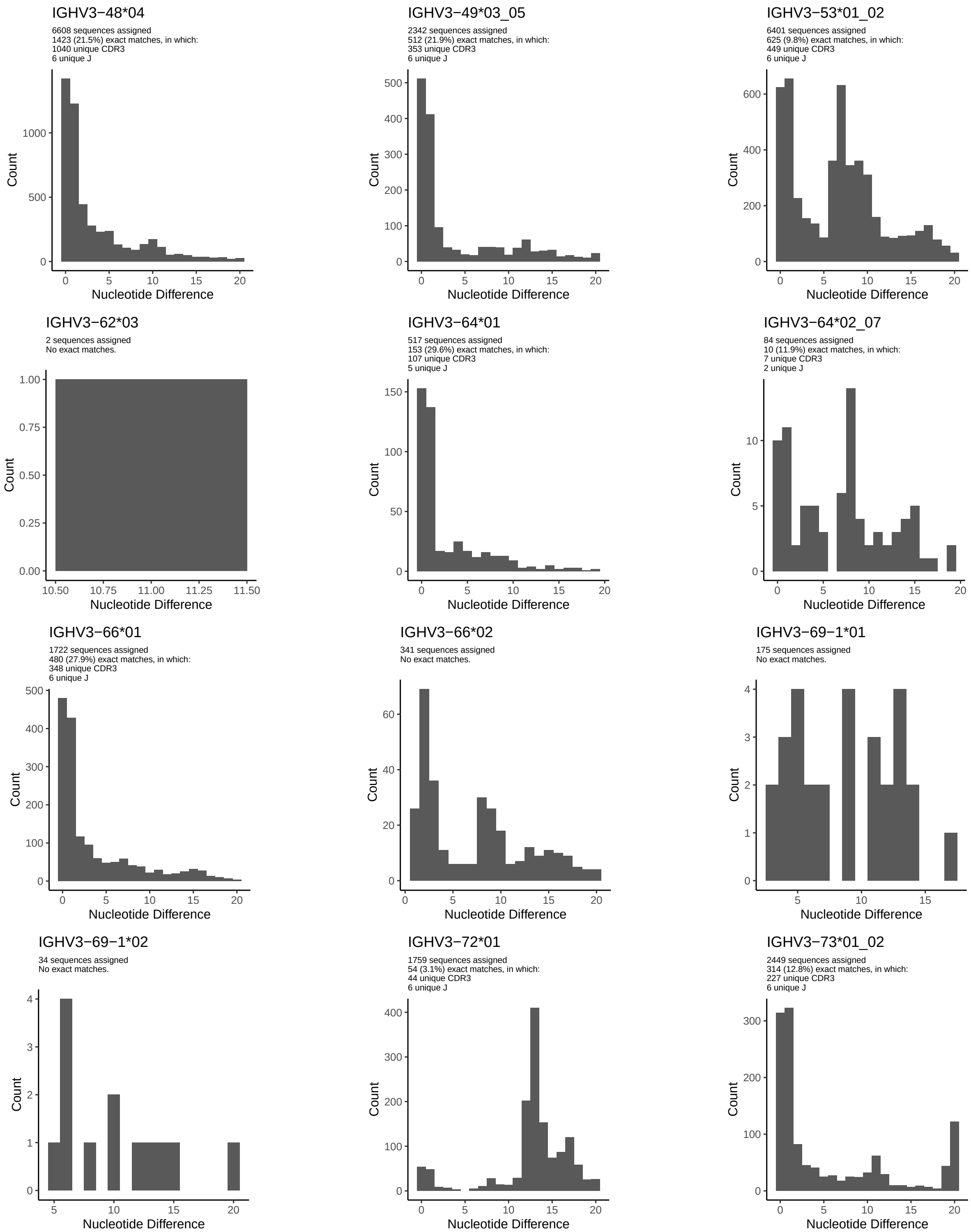
2 Variation from germline, in assignments to each allele

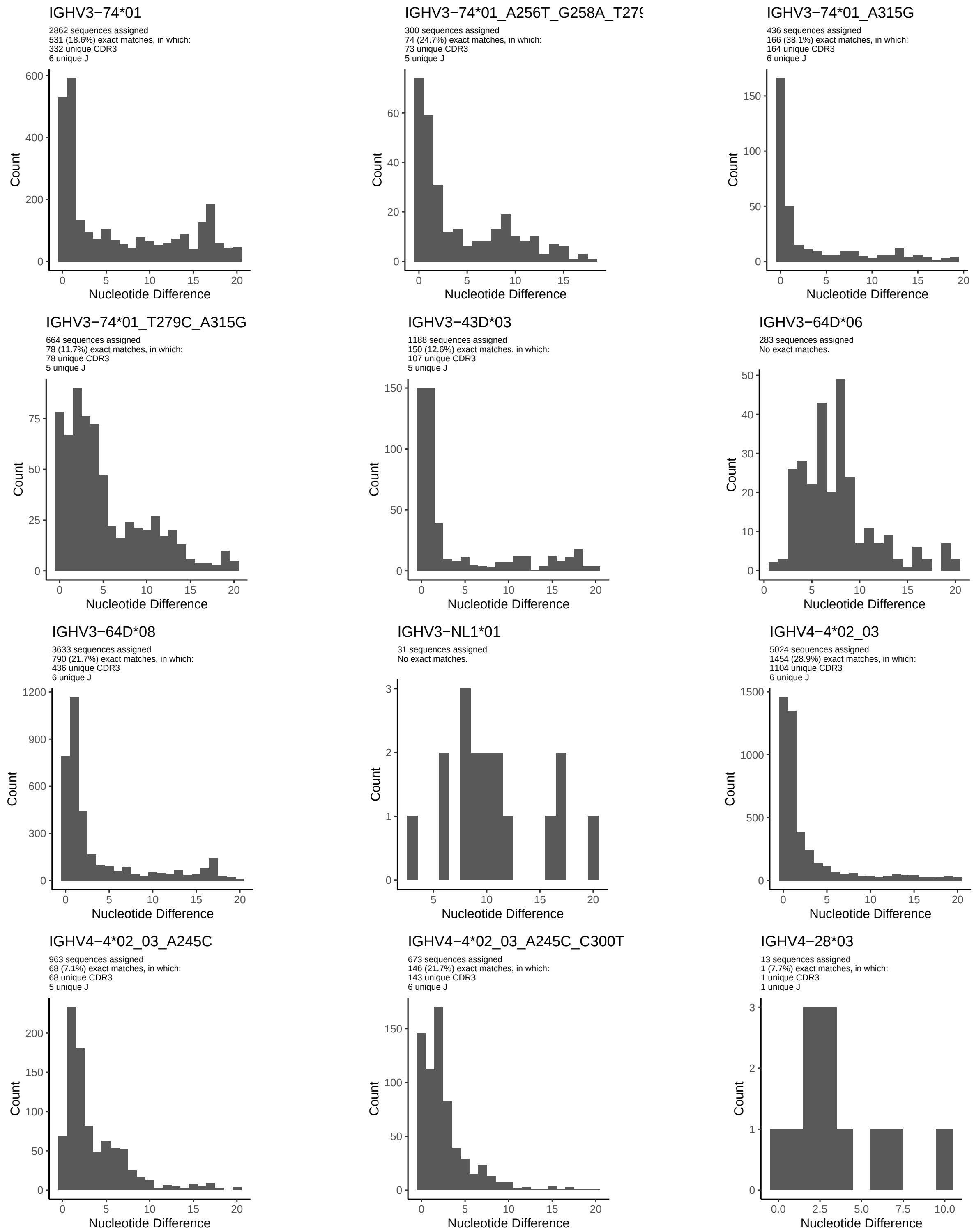


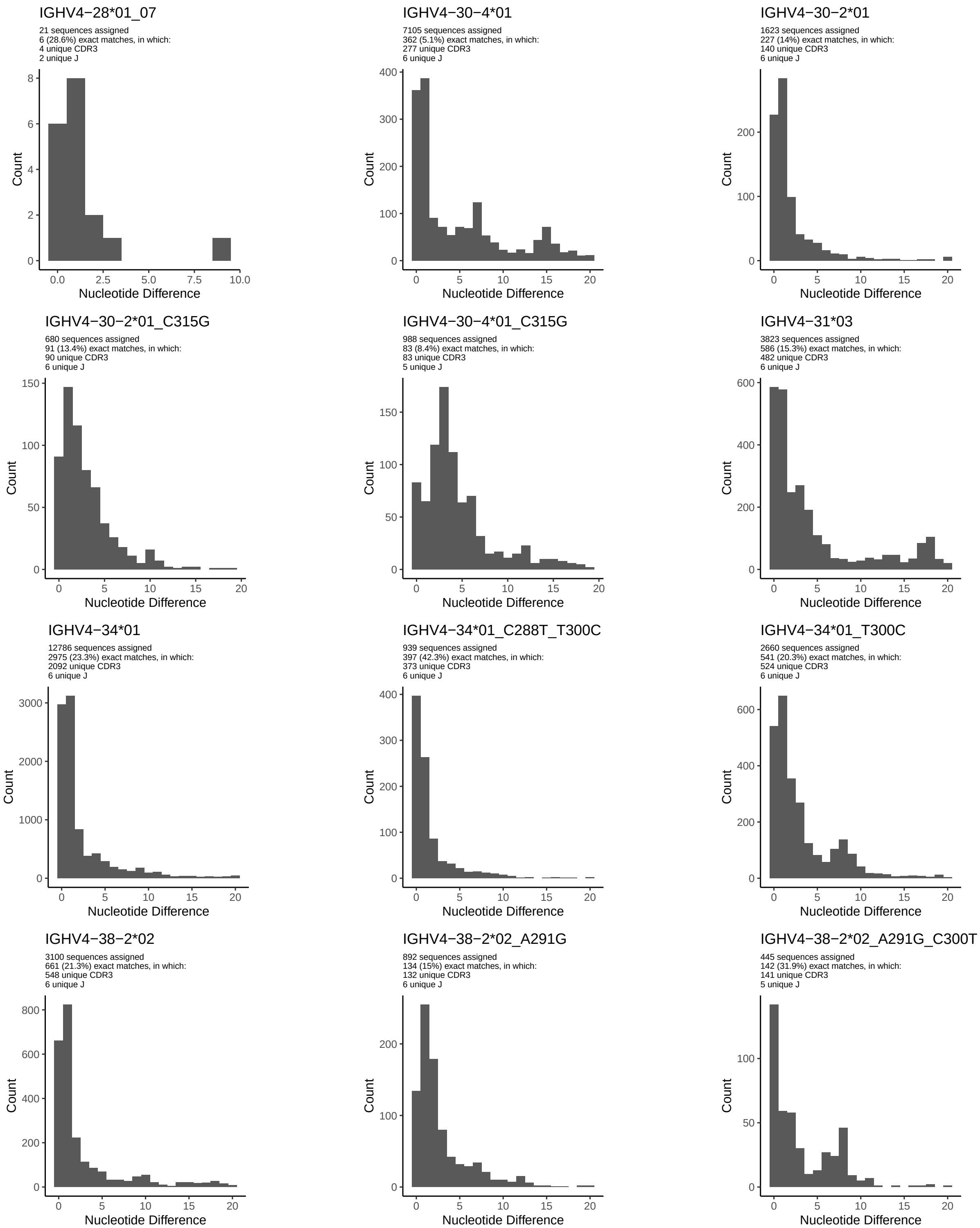

IGHV1−69*04_09
6418 sequences assigned
1868 (29.1%) exact matches, in which:
1516 unique CDR3
6 unique J
IGHV2−5*01
1057 sequences assigned
213 (20.2%) exact matches, in which:
174 unique CDR3
6 unique J
IGHV2−5*02
2275 sequences assigned
293 (12.9%) exact matches, in which:
224 unique CDR3
6 unique J
IGHV2−26*01
776 sequences assigned
196 (25.3%) exact matches, in which:
158 unique CDR3
5 unique J
IGHV2−70*01
1262 sequences assigned
135 (10.7%) exact matches, in which:
113 unique CDR3
5 unique J
IGHV2−70*15
632 sequences assigned
170 (26.9%) exact matches, in which:
132 unique CDR3
5 unique J
IGHV2−70D*04
1 sequences assigned
No exact matches.
IGHV3−7*01
5810 sequences assigned
950 (16.4%) exact matches, in which:
725 unique CDR3
6 unique J
IGHV3−7*03
1021 sequences assigned
94 (9.2%) exact matches, in which:
85 unique CDR3
5 unique J
IGHV3−7*05
3392 sequences assigned
499 (14.7%) exact matches, in which:
361 unique CDR3
6 unique J
IGHV3−9*01
4138 sequences assigned
1162 (28.1%) exact matches, in which:
786 unique CDR3
6 unique J
IGHV3−9*01_C258A_T279C_T288
1039 sequences assigned
160 (15.4%) exact matches, in which:
157 unique CDR3
6 unique J
Count
Nucleotide Difference

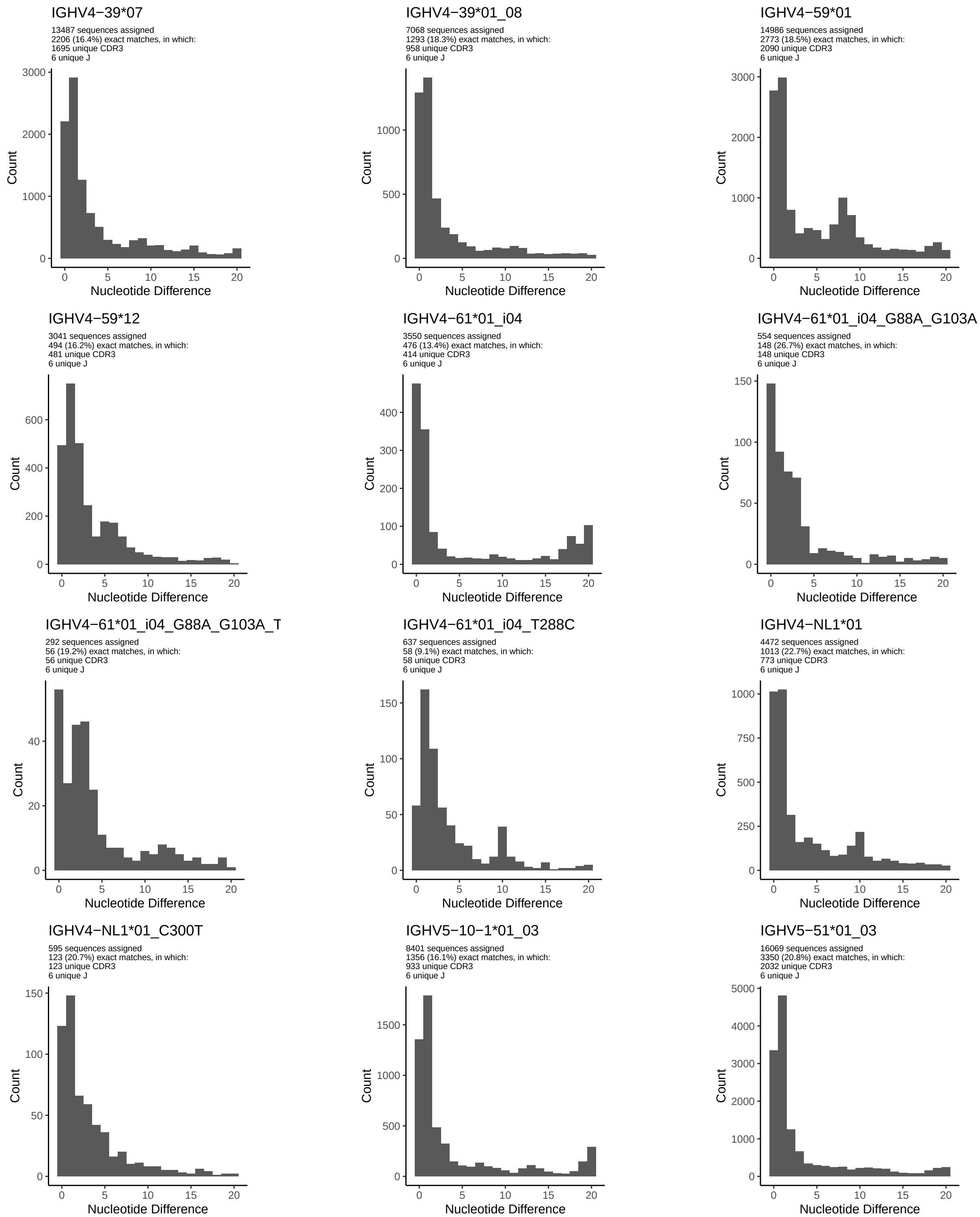

IGHV3−11*01
4569 sequences assigned
916 (20%) exact matches, in which:
675 unique CDR3
6 unique J
IGHV3−11*06
2181 sequences assigned
639 (29.3%) exact matches, in which:
454 unique CDR3
6 unique J
IGHV3−13*01
744 sequences assigned
235 (31.6%) exact matches, in which:
153 unique CDR3
5 unique J
IGHV3−13*05
154 sequences assigned
59 (38.3%) exact matches, in which:
44 unique CDR3
4 unique J
IGHV3−13*06
67 sequences assigned
4 (6%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV3−15*01
7278 sequences assigned
864 (11.9%) exact matches, in which:
618 unique CDR3
6 unique J
IGHV3−20*04
372 sequences assigned
2 (0.5%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−20*01_02
219 sequences assigned
44 (20.1%) exact matches, in which:
28 unique CDR3
6 unique J
IGHV3−21*01
13940 sequences assigned
3156 (22.6%) exact matches, in which:
2120 unique CDR3
6 unique J
IGHV3−23*01_04
28280 sequences assigned
4438 (15.7%) exact matches, in which:
2837 unique CDR3
6 unique J
IGHV3−23*01_04_C300T_A303G_
5091 sequences assigned
482 (9.5%) exact matches, in which:
467 unique CDR3
6 unique J
IGHV3−30−3*01
5804 sequences assigned
1698 (29.3%) exact matches, in which:
1237 unique CDR3
6 unique J
Count
Nucleotide Difference

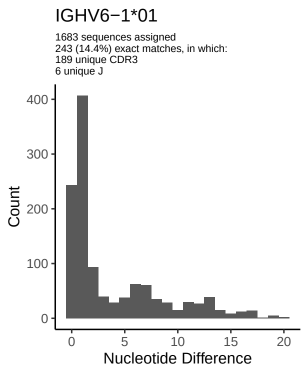

IGHV3−30*03
1785 sequences assigned
488 (27.3%) exact matches, in which:
465 unique CDR3
6 unique J
IGHV3−30*18
13262 sequences assigned
1946 (14.7%) exact matches, in which:
1316 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
1100 sequences assigned
162 (14.7%) exact matches, in which:
160 unique CDR3
6 unique J
IGHV3−30*03_T246C_T247G_A25
663 sequences assigned
105 (15.8%) exact matches, in which:
104 unique CDR3
5 unique J
IGHV3−30*03_T288C
1004 sequences assigned
286 (28.5%) exact matches, in which:
278 unique CDR3
6 unique J
IGHV3−33*01
3965 sequences assigned
891 (22.5%) exact matches, in which:
663 unique CDR3
6 unique J
IGHV3−33*06_C288T
417 sequences assigned
157 (37.6%) exact matches, in which:
152 unique CDR3
6 unique J
IGHV3−33*06_T300C_G303A
662 sequences assigned
125 (18.9%) exact matches, in which:
125 unique CDR3
6 unique J
IGHV3−38−3*01
52 sequences assigned
9 (17.3%) exact matches, in which:
9 unique CDR3
3 unique J
IGHV3−38*03
5 sequences assigned
No exact matches.
IGHV3−43*01
1393 sequences assigned
116 (8.3%) exact matches, in which:
82 unique CDR3
5 unique J
IGHV3−48*01
4336 sequences assigned
861 (19.9%) exact matches, in which:
602 unique CDR3
6 unique J
Count
Nucleotide Difference

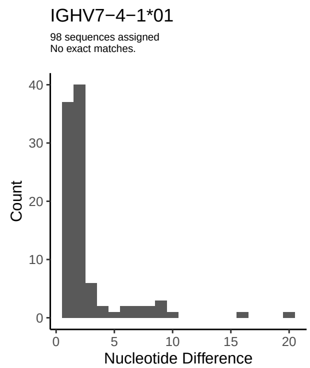

IGHV3−48*04
6608 sequences assigned
1423 (21.5%) exact matches, in which:
1040 unique CDR3
6 unique J
IGHV3−49*03_05
2342 sequences assigned
512 (21.9%) exact matches, in which:
353 unique CDR3
6 unique J
IGHV3−53*01_02
6401 sequences assigned
625 (9.8%) exact matches, in which:
449 unique CDR3
6 unique J
IGHV3−62*03
2 sequences assigned
No exact matches.
IGHV3−64*01
517 sequences assigned
153 (29.6%) exact matches, in which:
107 unique CDR3
5 unique J
IGHV3−64*02_07
84 sequences assigned
10 (11.9%) exact matches, in which:
7 unique CDR3
2 unique J
IGHV3−66*01
1722 sequences assigned
480 (27.9%) exact matches, in which:
348 unique CDR3
6 unique J
IGHV3−66*02
341 sequences assigned
No exact matches.
IGHV3−69−1*01
175 sequences assigned
No exact matches.
IGHV3−69−1*02
34 sequences assigned
No exact matches.
IGHV3−72*01
1759 sequences assigned
54 (3.1%) exact matches, in which:
44 unique CDR3
6 unique J
IGHV3−73*01_02
2449 sequences assigned
314 (12.8%) exact matches, in which:
227 unique CDR3
6 unique J
Count
Nucleotide Difference

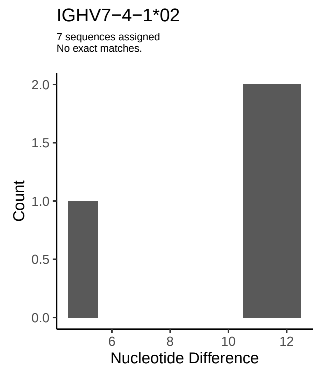

IGHV3−74*01
2862 sequences assigned
531 (18.6%) exact matches, in which:
332 unique CDR3
6 unique J
IGHV3−74*01_A256T_G258A_T279
300 sequences assigned
74 (24.7%) exact matches, in which:
73 unique CDR3
5 unique J
IGHV3−74*01_A315G
436 sequences assigned
166 (38.1%) exact matches, in which:
164 unique CDR3
6 unique J
IGHV3−74*01_T279C_A315G
664 sequences assigned
78 (11.7%) exact matches, in which:
78 unique CDR3
5 unique J
IGHV3−43D*03
1188 sequences assigned
150 (12.6%) exact matches, in which:
107 unique CDR3
5 unique J
IGHV3−64D*06
283 sequences assigned
No exact matches.
IGHV3−64D*08
3633 sequences assigned
790 (21.7%) exact matches, in which:
436 unique CDR3
6 unique J
IGHV3−NL1*01
31 sequences assigned
No exact matches.
IGHV4−4*02_03
5024 sequences assigned
1454 (28.9%) exact matches, in which:
1104 unique CDR3
6 unique J
IGHV4−4*02_03_A245C
963 sequences assigned
68 (7.1%) exact matches, in which:
68 unique CDR3
5 unique J
IGHV4−4*02_03_A245C_C300T
673 sequences assigned
146 (21.7%) exact matches, in which:
143 unique CDR3
6 unique J
IGHV4−28*03
13 sequences assigned
1 (7.7%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference


IGHV4−28*01_07
21 sequences assigned
6 (28.6%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV4−30−4*01
7105 sequences assigned
362 (5.1%) exact matches, in which:
277 unique CDR3
6 unique J
IGHV4−30−2*01
1623 sequences assigned
227 (14%) exact matches, in which:
140 unique CDR3
6 unique J
IGHV4−30−2*01_C315G
680 sequences assigned
91 (13.4%) exact matches, in which:
90 unique CDR3
6 unique J
IGHV4−30−4*01_C315G
988 sequences assigned
83 (8.4%) exact matches, in which:
83 unique CDR3
5 unique J
IGHV4−31*03
3823 sequences assigned
586 (15.3%) exact matches, in which:
482 unique CDR3
6 unique J
IGHV4−34*01
12786 sequences assigned
2975 (23.3%) exact matches, in which:
2092 unique CDR3
6 unique J
IGHV4−34*01_C288T_T300C
939 sequences assigned
397 (42.3%) exact matches, in which:
373 unique CDR3
6 unique J
IGHV4−34*01_T300C
2660 sequences assigned
541 (20.3%) exact matches, in which:
524 unique CDR3
6 unique J
IGHV4−38−2*02
3100 sequences assigned
661 (21.3%) exact matches, in which:
548 unique CDR3
6 unique J
IGHV4−38−2*02_A291G
892 sequences assigned
134 (15%) exact matches, in which:
132 unique CDR3
6 unique J
IGHV4−38−2*02_A291G_C300T
445 sequences assigned
142 (31.9%) exact matches, in which:
141 unique CDR3
5 unique J
Count
Nucleotide Difference

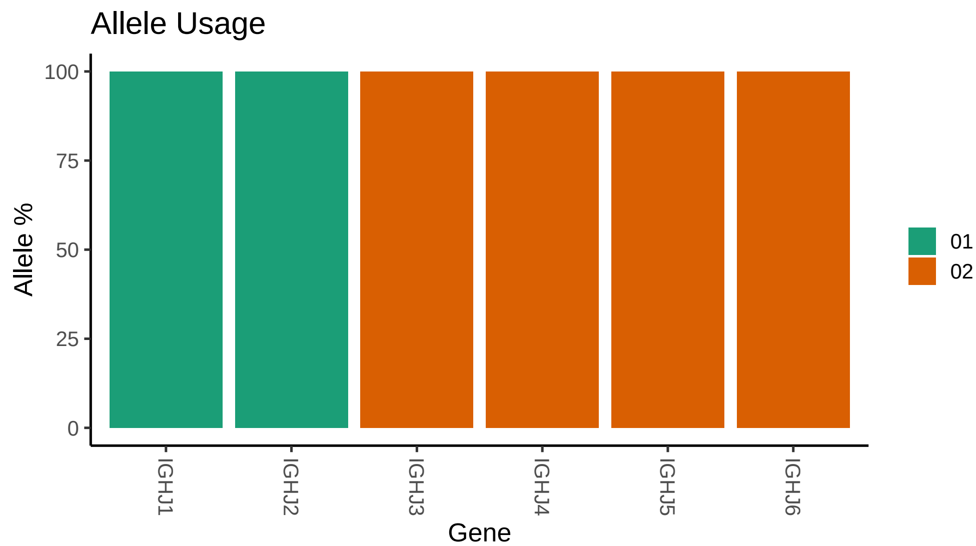

IGHV4−39*07
13487 sequences assigned
2206 (16.4%) exact matches, in which:
1695 unique CDR3
6 unique J
IGHV4−39*01_08
7068 sequences assigned
1293 (18.3%) exact matches, in which:
958 unique CDR3
6 unique J
IGHV4−59*01
14986 sequences assigned
2773 (18.5%) exact matches, in which:
2090 unique CDR3
6 unique J
IGHV4−59*12
3041 sequences assigned
494 (16.2%) exact matches, in which:
481 unique CDR3
6 unique J
IGHV4−61*01_i04
3550 sequences assigned
476 (13.4%) exact matches, in which:
414 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A
554 sequences assigned
148 (26.7%) exact matches, in which:
148 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A_T
292 sequences assigned
56 (19.2%) exact matches, in which:
56 unique CDR3
6 unique J
IGHV4−61*01_i04_T288C
637 sequences assigned
58 (9.1%) exact matches, in which:
58 unique CDR3
6 unique J
IGHV4−NL1*01
4472 sequences assigned
1013 (22.7%) exact matches, in which:
773 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
595 sequences assigned
123 (20.7%) exact matches, in which:
123 unique CDR3
6 unique J
IGHV5−10−1*01_03
8401 sequences assigned
1356 (16.1%) exact matches, in which:
933 unique CDR3
6 unique J
IGHV5−51*01_03
16069 sequences assigned
3350 (20.8%) exact matches, in which:
2032 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV6−1*01
1683 sequences assigned
243 (14.4%) exact matches, in which:
189 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV7−4−1*01
98 sequences assigned
No exact matches.
Count
Nucleotide Difference


IGHV7−4−1*02
7 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 109/M64 109/M64 109/M64 109/a8/b180434f6b24a6a4bdac73
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 109/M64 109/M64 109/M64 109/a8/b180434f6b24a6a
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 109/M64 109/M64 109/M64 109/a8/b180434f6b24a6a4bdac
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV3 9 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_C258A_T279C_T288C_C300T_T301G_A315G_A317G: replacing with
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T246C_T247G_A256T_G258A_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_C288T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 74 01_T279C_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A256T_G258A_T279C_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 01_i04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A_T288C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
```